gpeyt1-01 · SPT (N1)60

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A boring for a highway embankment encounters 8 ft of moist silty sand (γ = 112 pcf) over saturated clean fine sand (γsat = 124 pcf). The water table is at 8 ft. At a depth of 22 ft the field SPT blow count is N = 14 blows/ft. The automatic hammer was calibrated at an energy ratio of 80%. The driller also reports a 2 ft sample recovery, a 140 lb hammer with a 30 in. drop, and AWJ rods (none of which need further correction for this problem).

Most nearly, the corrected blow count (N1)60 at 22 ft is:

A. 20
B. 17
C. 15
D. 11

Given equations / data

> γw = 62.4 pcf; N60 = (Eeff/60)·Nmeas; (N1)60 = CN·N60; CN = 0.77 log10(20/Po), Po = vertical effective stress in tsf

Worked solution

1. Po = [8(112) + 14(124 − 62.4)]/2000 = 0.879 tsf
2. N60 = 14×80/60 = 18.67
3. CN = 0.77 log10(20/0.879) = 1.045
4. (N1)60 = 18.67×1.045 = 19.5 ≈ 20

Final answer

20

Why the other values are wrong

- **17** — used total stress for Po
- **15** — skipped the energy correction
- **11** — inverted the energy ratio (60/80)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2 (Hammer Efficiency, Depth Adjustment), p. 124

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwQ24-zlxyU, https://www.youtube.com/watch?v=DJaHlgyaOAg (no creator values copied).

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